

PEDIATRICS®

OFFICIAL JOURNAL OF THE AMERICAN ACADEMY OF PEDIATRICS

United Kingdom Collaborative Randomized Trial of Neonatal Extracorporeal Membrane Oxygenation: Follow-up to Age 7 Years

Helena McNally, Charlotte C. Bennett, Diana Elbourne, David J. Field and for the UK Collaborative ECMO Trial Group

Pediatrics 2006;117:e845-e854; originally published online Apr 24, 2006;

DOI: 10.1542/peds.2005-1167

The online version of this article, along with updated information and services, is located on the World Wide Web at:

<http://www.pediatrics.org/cgi/content/full/117/5/e845>

PEDIATRICS is the official journal of the American Academy of Pediatrics. A monthly publication, it has been published continuously since 1948. PEDIATRICS is owned, published, and trademarked by the American Academy of Pediatrics, 141 Northwest Point Boulevard, Elk Grove Village, Illinois, 60007. Copyright © 2006 by the American Academy of Pediatrics. All rights reserved. Print ISSN: 0031-4005. Online ISSN: 1098-4275.

American Academy of Pediatrics

DEDICATED TO THE HEALTH OF ALL CHILDREN™



United Kingdom Collaborative Randomized Trial of Neonatal Extracorporeal Membrane Oxygenation: Follow-up to Age 7 Years

Helena McNally, MSc^a, Charlotte C. Bennett, MD^b, Diana Elbourne, PhD^c, David J. Field, MD^d, for the UK Collaborative ECMO Trial Group

^aMedical Sciences Division, University of Oxford; ^bNeonatal Unit, John Radcliffe Hospital, Oxford, United Kingdom; ^cMedical Statistics Unit, London School of Hygiene and Tropical Medicine, Keppel Street, London, United Kingdom; ^dNeonatal Unit, Leicester Royal Infirmary, Leicester, United Kingdom

The authors have indicated they have no financial relationships relevant to this article to disclose.

ABSTRACT

OBJECTIVE. The UK Collaborative ECMO trial provided an opportunity to describe mortality and morbidity associated with a neonatal ECMO policy compared with conventional management. The improved survival in the ECMO group was not offset by an increase in disability at 4 years, but the children were too young to assess educational and other longer-term impacts. The objective of this study was to assess the longer-term impact of these policies at age 7 years.

METHODS. A psychologist assessed 90 of the 100 children available for follow-up without prior knowledge of treatment allocation. The assessments took place at the children's schools within 3 months of their 7th birthdays.

RESULTS. Sixty-eight of 89 (76%) children recorded a cognitive level within the normal range. Learning problems were similar in the 2 groups, and there were notable difficulties with spatial and processing tasks. A higher respiratory morbidity and increased risk of behavioral problems among children treated conventionally persisted. Progressive sensorineural hearing loss was found in both groups.

CONCLUSIONS. The underlying disease processes appear to be the major influence on morbidity at 7 years. The beneficial influence of an ECMO policy is still present at 7 years.

www.pediatrics.org/cgi/doi/10.1542/peds.2005-1167

doi:10.1542/peds.2005-1167

Key Words

follow-up studies, respiratory distress, neonatal care, long-term survival, long-term follow-up

Abbreviations

CI—confidence interval
BAS—British Ability Scales 195
GCAS—General Conceptual Ability Score
MABC—Movement Assessment Battery for Children
SD—standard deviation
SDQ—Strengths and Difficulties Questionnaire
SNHL—sensorineural hearing loss

Accepted for publication Nov 7, 2005

Address correspondence to Helena McNally, MSc, Medical Sciences Division, University of Oxford, John Radcliffe Hospital, Oxford, OX3 9DU, United Kingdom; E-mail: mcnally@medschool.ox.ac.uk

PEDIATRICS (ISSN Numbers: Print, 0031-4005; Online, 1098-4275). Copyright © 2006 by the American Academy of Pediatrics

THE UK COLLABORATIVE ECMO Trial randomized 185 infants with severe respiratory failure meeting standard ECMO criteria to a policy of transfer to 1 of 5 ECMO centers or to receive conventional neonatal intensive care, usually in the local level 3 intensive care unit. The trial demonstrated that use of ECMO in this way was clinically and cost-effective to the age of 1 year.¹⁻³ The primary outcome of known death or severe disability by age 4 years was 34 of 93 (37%) in the ECMO group versus 54 of 92 (59%) in the conventional treatment group; relative risk 0.64 (95% confidence interval [CI] 0.47–0.86); $P = .004$.³ These results are consistent with the systematic review of all the trials of neonatal ECMO.⁴

Infants recruited to the study were critically ill and considered to be at high risk of disability. Although a “normal survival” advantage had been clearly demonstrated in terms of performance at 4 years, it is generally accepted that the true extent of acquired disability among survivors can change over time. Such an effect was clearly seen when comparing the outcome of children at 1 and 4 years when a clearer assessment of cognitive skills, coordination, behavioral difficulties, and sensory loss became possible. There were concerns that 60% of children in both randomized groups had abnormal neuromotor signs and this particularly affected the execution of fine motor tasks. Respiratory morbidity and behavioral problems were more frequently seen in the conventionally managed group, and a small number of children in both randomized groups had developed progressive severe sensorineural deafness.

It could be anticipated that some of these earlier difficulties would have considerable bearing on a child's progress; whereas other problems might reduce in impact over time. The present study aimed to evaluate the impact of the neonatal policies as the children started elementary school around the age of 7 years.

METHODS

The families recruited to the UK ECMO Trial had been contacted annually and were aware that, with their consent, a further assessment was planned at 7 years of age. Before their 7th birthday, their general practitioner was contacted by the Trial coordinator to ensure that continued involvement was considered appropriate. If this confirmation was forthcoming, a letter was sent to the family requesting permission to arrange a follow-up assessment at the child's school. Permission of the head teacher was also sought and the Local Education Authority was informed. Assessments took place within 3 months of the child's 7th birthday and were undertaken by a single developmental psychologist (HMcN). She had no prior knowledge of the child's trial allocation, disease course, or their previous assessment findings. The children usually wore a “polo shirt,” which hid any scar. Parents (and children) were also asked not to reveal the allocation before or during the assessment. Parents al-

ways complied with this request, although in 3 instances, a child told the assessor that they had had an operation when they were an infant.

The parents, general practitioner, and class teacher were asked to complete standardized questionnaires recording the child's behavior,⁵ ability in class, communication skills,⁶ and the use of health service resources. In addition, the child's primary caregiver was asked to complete the General Health Questionnaire.⁷ The psychologist completed a predefined and standardized assessment to determine outcome status in each of 6 clinical domains (cognitive ability, neuromotor skills, general health, behavior, hearing, and vision). In one case, it was known that the child was severely disabled and required a qualitative assessment.

Cognitive ability was assessed using the British Ability Scales II (BAS),⁸ including cluster scores of verbal, non-verbal reasoning, and spatial abilities. BAS II achievement scales were used to assess number skills, spelling, and word reading. Reading comprehension was tested using a scale from WORD.⁹ Visual memory was assessed using the Rey Complex Figure Test¹⁰ and verbal memory using the Children's Memory Scale.¹¹ The summary score for the BAS scale is the general conceptual ability score (GCAS). A score <50 was considered severe cognitive disability, 50 to 69 moderate, and 70 to 84 mild disability. Children with GCAS above 84 but a BAS cluster or achievement score <10th percentile was classified as having cognitive impairment. All other children were classified as normal. Information was collected about the child's schooling and whether they had been recorded as having special educational needs. The class teacher was asked to indicate the children's ability with respect to a typical mixed ability class using a visual analog scale from poor to excellent. They also completed the Children's Communication Checklist.⁶

Neuromotor performance was measured using components of the Movement Assessment Battery for Children (MABC).¹² This assessed fine motor skills and balance. When gross motor difficulties were observed, these were also recorded. Children who scored within 1 standard deviation (SD) of the mean for their age on all subtests administered were deemed to have a normal level of neuromotor development, between 1 and 2 SDs on 1 or 2 subtests were classified as having an impairment of neuromotor development; consistent scoring between 1 and 2 SDs was classified as having a mild disability, scoring >2 SDs below the mean was classified as moderate disability, and inability to participate in any of the tests because of neuromotor difficulties was classified as severe disability.

The detailed questionnaire to parents about the child's general health and use of health services and special educational services specifically asked about the presence of fits, asthma, and the requirement for regular medication or hospital readmission. The economic im-

plications of this are reported in the accompanying article.¹³ Children without a recognized illness were deemed to have “normal” general health. Those who had a chronic illness that was well controlled by medication taken regularly, and had required hospital admission less frequently than once each 6 months between ages 4 to 7 years, were deemed to have general health impairment only. Those children who had an ongoing chronic illness and who required frequent hospital admission or close supervision were considered to be disabled in terms of their general health.

To ascertain behavior, the child’s parent and teacher were asked to complete the Strengths and Difficulties Questionnaire (SDQ).⁵ This score has been developed to anticipate that 10% of children assessed will have a score within the “abnormal” range. The children were classified to have a behavioral disability if the difficulties were considered by the parent to adversely impact on their child’s daily living. If this required specialist intervention, the children were classified to have a moderate behavioral disability.

A pure-tone audiometry profile was completed during the assessment with additional information obtained from hospital records and parental reports if required. Information about the child’s vision was obtained from parental report.

The outcome classification in each domain is illustrated in Table 1.

Overall outcome was described as normal, impaired, or disabled by the degree of functional loss over these domains. Children with a test score outside of the normal range were defined as impaired; those with some functional loss but who need little additional support were regarded as having a mild disability; children who used aids and required assistance were classified as moderately disabled; and children requiring constant supervision were defined as severely disabled. A child’s overall status was allocated to the highest degree of disability in any of the 6 clinical domains. These outcome categories were directly comparable to those used to classify children after their assessment at age 4 years and in keeping with recommended outcome classification systems for school-aged children.¹⁴ Children who were unable to carry out an assessment because of poor ability were given the lowest score except in timed tests when they were recorded as unknown (Rey Complex Figure and MABC).^{10,11}

Research ethics approval was sought from local committees in anticipation of the 7-year follow-up. Data were double-entered and analyzed by intention to treat using SPSS for windows (version 11.5). Differences between the randomized groups are presented as relative risks, risk differences, or differences between means, as appropriate, with 95% confidence intervals. Children

lost to follow-up were assumed to be alive and without severe disability, and sensitivity analyses were used to check the robustness of this assumption.

RESULTS

Ninety children were assessed at age 7 years (Fig 1). No additional deaths were known to have occurred between the 4- and 7-year follow-up assessments.

Ten children (10%) were lost to follow-up from the time of discharge from the hospital (6 ECMO, 4 conventional). Information describing 6 children lost up to age 4 years have been previously described. Of the 4 children who were lost to follow-up between ages 4 and 7 years, 2 (both ECMO) were considered to have a mild disability when seen at age 4 years and 2 (1 ECMO, 1 conventional) were classified with moderate disability because of cognitive problems.

All assessed children were seen between the ages of 7 years and 7 years 3 months. The characteristics at the time of trial entry for those infants recruited and of the survivors assessed at age 7 years are shown in Table 2. One ECMO child did not complete a full psychologic assessment because the family was living abroad, but parent and teacher questionnaires were returned. Of the 56 surviving children seen at 7 years of age who were randomized to an ECMO policy, 50 (89%) actually received ECMO support (31 venoarterial cannulation [VA], 16 venovenous [VV], and 3 both VA and VV).

Overall classifications in each of the clinical domains is indicated in Table 3. The information on which they are based is described in the following text and in Tables 4 through 8.

Cognitive

At age 7 years, like at 4 years, there was no evidence of a difference between the 2 trial groups. The mean GCAS was similar in the 2 groups (ECMO 95.4 versus conventional 96.0). Encouragingly, 68 of 89 (76% of the children doing the BAS test in either group) recorded a cognitive level within the normal range. There was no evidence of differences between the randomized groups for mean cluster scores of verbal, nonverbal reasoning, spatial abilities, or special nonverbal composite, although these generally fell below population norms. The children had most difficulty undertaking tasks of spatial ability such as pattern construction and recall of designs, with 23 of 89 (26%) scoring below the 10th percentile. There was no evidence of a difference in the mean BAS achievement scales between the 2 randomized groups, and these results were generally higher on average than the ability scores. However, the children also had particular difficulty completing tasks of reading comprehension with 35 of 89 (39%) scoring below the 10th percentile (Table 4).

Four of the 55 children (all ECMO) did not complete the visual memory assessment (3 unable, 1 refusal).

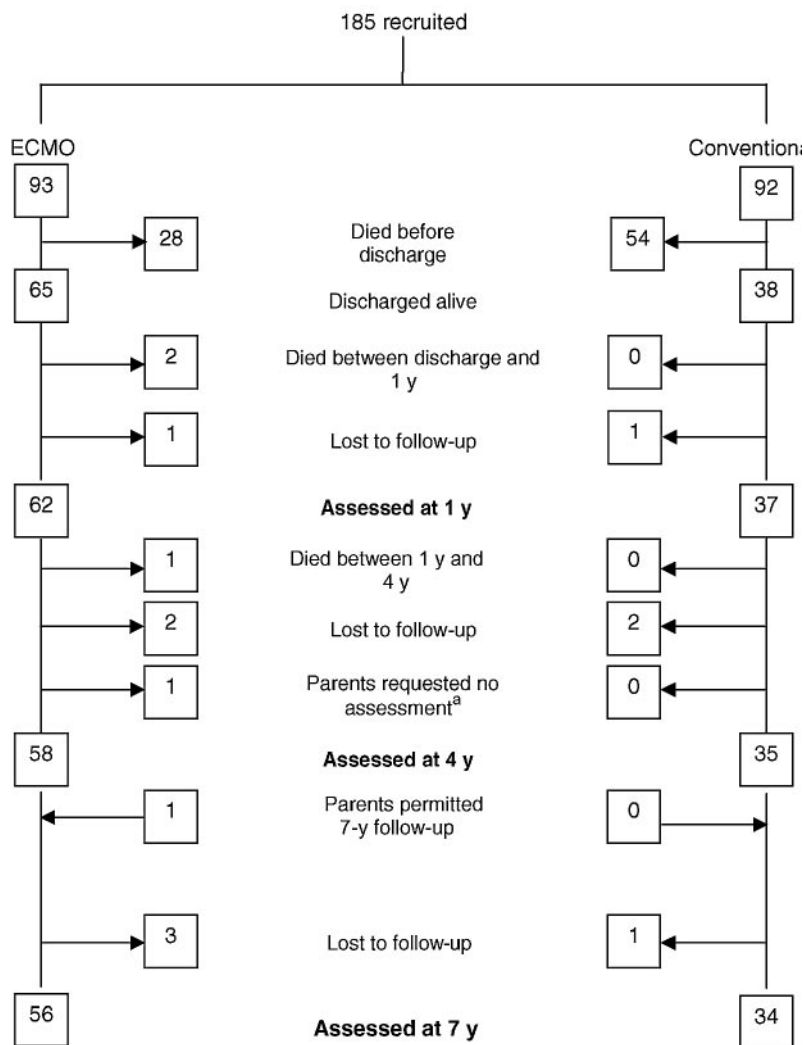
TABLE 1 Outcome Classification at 7-Year Assessment: UK Collaborative ECMO Trial

	Cognitive	Neuromotor	General Health	Behavior	Vision	Hearing
Impairment "Problems in body structure with no loss of function"	Specific learning difficulty and GCAS ≥ 85 , < 10 th percentile for any BAS cluster score	Abnormal signs on neurologic examination but normal function Score < -1 SD on 1–2 MABC subtests	Medical condition that requires medication most days	Abnormal score on any of the 5 subsections on SDQ	Abnormal eye movements or squint; corrected refractive error with VA 6/6	Unilateral hearing with normal hearing in good ear
Mild disability "Some loss of function but able to function independently"	GCAS 70–84 (< 1 SD)	Score < -1 SD on all MABC subtests; walks without restriction; limitations in balance and coordination only	Chronic medical condition requiring > 1 admission in 6 months; epilepsy with > 1 fit/month	Abnormal total score on SDQ and abnormal impact score	VA in best eye (with refractive errors corrected) $< 6/6$ and $\geq 6/18$	Normally wears hearing aids
Moderate disability "Aids or assistance; moderate difficulty performing activities"	GCAS 50–69 (< -2 SD)	Score < -2 SD on all MABC subtests; walks with assistive devices; able to sit independently	Continued need for stoma or intensive care admission	Behavioral disturbance requiring supervision more than half the time or specialist intervention	Visual acuity $< 6/18$ and $\geq 6/60$	Normally wears aids; corrected loss still cause problems with schoolwork and difficulties with conversation
Severe disability "Constant supervision" "dependent on caregiver"	GCAS < 50 (< -3 SD)	Unable to perform MABC subtests; self-mobility with limitation or no means of independent mobility	Need for continuous home oxygen; frequent seizures—difficult to control	Behavioral disturbance requiring constant supervision	VA $< 6/60$, gross movement/light and dark only	Severe hearing loss not fully corrected with aids causing limitation of function and problems with hearing most of the time

VA indicates visual acuity.

FIGURE 1

Flow diagram of children recruited to the UK ECMO Trial to 7-year follow-up. ^aBefore 4-year follow-up, the child had recently had an intercurrent illness and parents felt a further assessment would be inappropriate at this time. Permission for 7-year follow-up was given.



Copying the figure, immediate and delayed recall was poor in both groups, especially among those children treated conventionally (Table 5). Children in both groups performed poorly in assessment of verbal memory. This was most notable for recall.

Overall, 19 (21%) children received special support in school (11 of 56 ECMO, 8 of 34 conventional) and 15 were recorded as having special educational needs (11 of 56 ECMO, 4 of 34 conventional). Teacher report describing the children's performance compared with a typical mixed-ability class was recorded for 86 children. There was no evidence of a difference in these scores between the 2 groups. Mean scores were below average in many areas of ability, notably creative writing and concentration span, and above average for cooperation in class. Scored using the Children's Communication Checklist, 9% of children in both groups were considered by their teachers to have communication difficulties that were likely to interfere with learning.

Nine children (4 of 55 [7%] ECMO versus 5 of 34 [15%] conventional) were classified with mild disability,

and 9 children (6 of 55 [11%] ECMO versus 3 of 34 [9%] conventional) were considered to have a moderate disability. Three children (all ECMO) were severely disabled.

Neuromotor

There was no evidence of difference in laterality between the 2 groups. Only 1 ECMO-allocated child demonstrated no preference in laterality.

Based on the MABC, 36 children (40%) were deemed to have a normal level of neuromotor development. Thirty-nine children (22 of 55 [39%] ECMO versus 17 of 34 [50%] conventional) were classified as having an impairment of neuromotor development. An additional 9 (4 of 55 [7%] ECMO versus 5 of 34 [18%] conventional) were classified as having a mild disability. Four children in the ECMO group were described as moderately disabled and 1 as severely disabled (an ECMO child who could not participate in any of the tests because of neuromotor difficulties) (Table 3).

TABLE 2 Characteristics of Babies at Trial Entry for the Total Sample and for Those in the 7-Year Follow-Up

	Total Sample		In 7-Yr Follow-up	
	ECMO (n = 93)	CM (n = 92)	ECMO (n = 56)	CM (n = 34)
Primary diagnosis at trial entry; no. (%)				
Congenital diaphragmatic hernia	18 (19)	17 (18)	3 (5)	0
Isolated persistent fetal circulation	15 (16)	16 (17)	11 (20)	6 (18)
Persistent pulmonary hypertension resulting from:	57 (61)	58 (63)	41 (73)	27 (79)
Meconium aspiration	32 (34)	37 (40)	25 (45)	17 (50)
Sepsis	14 (15)	5 (5)	8 (14)	1 (3)
Respiratory distress syndrome	8 (9)	12 (13)	5 (9)	7 (21)
Other	3 (3)	4 (4)	3 (5)	2 (6)
Other	3 (3)	1 (1)	1 (2)	1 (3)
Trial entry criteria; no. (%)				
Oxygen index 40–59	55 (59)	55 (60)	37 (66)	22 (65)
Oxygen index ≥ 60	37 (40)	37 (40)	18 (32)	12 (35)
PaCO ₂ >12 kPa for >3 hrs	1 (1)	0	1 (2)	0
Age at trial entry; no. (%)				
0 days	37 (40)	42 (46)	23 (41)	17 (50)
1 day	31 (33)	20 (22)	14 (25)	7 (21)
2–27 days	25 (27)	30 (33)	19 (34)	10 (29)
Gender; no. (%)				
Male	54 (58)	58 (63)	35 (63)	20 (59)
Female	39 (42)	34 (37)	21 (38)	14 (41)
Gestational age at delivery (in completed weeks)				
Median	39	40	39	39
Interquartile range	37–40	38–41	37–41	38–41
Birth weight (g); mean	3261	3346	3417	3583
(Standard deviation)	(516)	(593)	(481)	(615)
ECMO cannulation				
Venoarterial	50 (54)	0	31 (56)	0
Venovenous	23 (25)	1 (1)	16 (29)	1 (1)
Both	5 (5)	0	3 (5)	0
Neither	15 (16)	91 (99)	6 (10)	33 (99)
Days of high-pressure ventilation ^a before trial entry; no. (%)				
0	44 (47)	46 (50)	25 (45)	18 (53)
1	29 (31)	20 (22)	16 (29)	7 (21)
2–9	20 (22)	26 (28)	15 (27)	9 (26)
Use before trial entry of; no. (%)				
Any pulmonary vasodilators	80 (86)	78 (85)	48 (86)	25 (74)
Nitric oxide	13 (14)	11 (12)	6 (11)	4 (12)
Inotropes	75 (81)	71 (77)	46 (82)	26 (76)
High-frequency ventilation	13 (14)	14 (15)	8 (14)	1 (3)
Muscle relaxants	81 (87)	81 (88)	52 (93)	32 (94)
Surfactant	28 (30)	37 (40)	15 (27)	13 (38)
ECMO center (in which cot reserved); no. (%)				
Leicester	29 (31)	35 (38)	19 (34)	19 (56)
GOS, London	30 (32)	26 (28)	16 (29)	5 (15)
Newcastle	14 (15)	11 (12)	8 (14)	2 (6)
Glasgow	17 (18)	17 (18)	11 (20)	7 (21)
KCH, London	3 (3)	3 (3)	2 (4)	1 (3)

^a 25 cm H₂O peak pressure.

General Health

A higher respiratory morbidity was documented in children treated conventionally compared with those in the ECMO group. In the conventionally treated group, 11 of 30 (32%) continued to have intermittent attacks of wheezing in the 12 months before the assessment and 14 of 32 (41%) regularly used an inhaler. These compared with 6 of 43 (11%) ECMO children who wheezed and 14 of 48 (25%) using an inhaler. Six children (4

ECMO versus 2 conventional) had required hospital re-admission for respiratory reasons since assessment at 4 years of age (Table 6).

This respiratory morbidity contributed to an increased proportion of children treated conventionally who were classified to have an impairment in the general health domain at 7 years compared with those children in the ECMO group (12 of 54 [22%] ECMO versus 19 of 34 [56%] conventional). In the ECMO group, 1 child who

TABLE 3 Outcome Classification at 7-Year Assessment: UK Collaborative ECMO Trial

	Cognitive		Neuromotor		General Health		Behavior		Vision		Hearing	
	ECMO (n = 56) (%)	CM (n = 34) (%)	ECMO (n = 56) (%)	CM (n = 34) (%)	ECMO (n = 56) (%)	CM (n = 34) (%)	ECMO (n = 56) (%)	CM (n = 34) (%)	ECMO (n = 56) (%)	CM (n = 34) (%)	ECMO (n = 56) (%)	CM (n = 34) (%)
Not known	1 (2)	0	1 (2)	0	2 (4)	0	2 (4)	0	2 (4)	0	1 (2)	0
Normal	38 (68)	24 (70)	24 (43)	12 (35)	40 (71)	15 (44)	32 (57)	14 (41)	45 (80)	27 (80)	39 (70)	25 (74)
impairment	4 (7)	2 (6)	22 (39)	17 (50)	12 (21)	19 (56)	10 (18)	8 (24)	2 (4)	5 (15)	13 (23)	5 (15)
Mild disability	4 (7)	5 (15)	4 (7)	5 (15)	1 (2)	0	7 (12)	8 (24)	5 (9)	2 (6)	0	3 (9)
Moderate disability	6 (11)	3 (9)	4 (7)	0	0	0	4 (7)	4 (12)	1 (2)	0	3 (5)	1 (2)
Severe disability	3 (5)	0	1 (2)	0	1 (2)	0	1 (2)	0	1 (2)	0	0	0

ECMO indicates randomized to a policy of transfer to a UK ECMO center; CM: randomized to receive conventional neonatal intensive care.

TABLE 4 Cognitive Function

	ECMO (n = 56) ^a		CM (n = 34)	
	Mean Standard Score (SD)	No. <10th Percentile (%)	Mean Standard Score (SD)	No. <10th Percentile (%)
BAS cluster scores				
Spatial ability	93.95 (22.432)	16 (29)	92.29 (18.843)	7 (21)
Verbal ability	96.45 (17.103)	8 (14)	98.79 (14.936)	4 (12)
Nonverbal reasoning	97.85 (18.060)	8 (14)	99.24 (17.715)	8 (24)
Special nonverbal composite	95.47 (21.767)	13 (23)	95.15 (18.614)	8 (24)
BAS general conceptual ability score	95.35 (21.628)		96.00 (18.211)	
BAS achievement scales				
Number skills	101.04 (17.907)	9 (16)	101.26 (19.212)	5 (15)
Spelling	99.13 (20.931)	9 (16)	98.85 (21.211)	5 (15)
Word reading	98.69 (20.651)	8 (14)	96.82 (19.935)	5 (15)
Word				
Reading comprehension	89.84 (14.897)	22 (39)	85.85 (18.725)	13 (38)

^a One child allocated to ECMO was not available to carry out the cognitive assessment.

TABLE 5 Visual and Verbal Memory

		ECMO (n = 56)	CM (n = 34)
Rey Complex Figure Test			
Copy	No. ≤10th percentile (%) (not known)	27 (48) (5)	24 (71) (0)
Immediate recall	Mean t-score [SD] no. <10th percentile (%) (not known)	39.580 [10.6581] 23 (41) (6)	37.471 [9.359] 20 (59) (0)
Delayed recall	Mean t-score [SD] no. <10th percentile (%) (not known)	39.240 [11.6996] 25 (45) (6)	36.676 [10.1645] 22 (65) (0)
Children's Memory Scale			
Immediate recall	Mean standard score [SD] no. <10th percentile (%) (not known)	82.76 [18.781] 24 (43) (1)	82.88 [20.353] 11 (32) (0)
Delayed recall	Mean standard score [SD] no. <10th percentile (%) (not known)	84.62 [20.264] 20 (36) (1)	85.88 [23.862] 9 (27) (0)
Delayed recognition	Mean standard score [SD] no. <10th percentile (%) (not known)	93.07 [22.405] 14 (25) (1)	93.62 [23.873] 6 (18) (0)

had required hospital treatment on 5 or more occasions since the age of 4 years was considered to have a mild disability and another 1 child classified with severe disability had frequent seizures that were difficult to control (Table 3).

Behavior

Behavior problems reported by a parent were more common in the conventionally treated group. Parents also

reported fewer prosocial or positive behaviors in conventionally treated children (Table 7). The most commonly described difficulty was hyperactivity. Total deviance score was higher in the conventionally treated group (38% versus 18% ECMO). Teacher-reported scores demonstrated that a higher proportion of children in the conventionally treated group experienced behavioral difficulties in all areas assessed. Overall, 22 of 85 (26%) children had difficulties with hyperactivity.

TABLE 6 General Health of Child at 7 Years

	ECMO (n = 56)	CM (n = 34)
Since birth: no. (%) (not known)		
Asthma diagnosed in last 12 months: no. (%) (not known)	16 (29) (10)	18 (53) (5)
Attacks of wheezing	6 (11) (13)	11 (32) (4)
Inhaler used	14 (25) (8)	14 (41) (2)
Since 4 years old: no. (%) (not known)		
Other medical problem	16 (29) (4)	16 (47) (0)
On regular medication	16 (29) (2)	14 (41) (0)
Admission to hospital for respiratory problems	4 (7) (1)	2 (6) (0)
Seizures/fits	3 (5) (3)	1 (3) (0)
Funny turns ^a	6 (11) (2)	5 (15) (0)
On medication for seizures/funny turns	3 (5) (3)	2 (6) (0)

^a A lay term generally synonymous with fits but often better understood by parents.

Behavioral disability was more commonly described among those children treated conventionally (12 of 52 [23%] ECMO versus 12 of 34 [36%] conventional).

Eight of these children (4 of 54 [7%] ECMO versus 4 of 34 [12%] conventional) had required specialist intervention for their behavioral difficulties and these children were classified as having a moderate disability. One child treated with ECMO had severe behavioral problems requiring constant supervision (Table 3).

Hearing and Vision

Five children had hearing aids (2 ECMO, 3 conventional). Two children were identified as having worsening sensorineural hearing loss between the ages of 4 and 7 years. Four children (3 ECMO, 1 CM) had a significant hearing loss. This was uncorrected, despite aids, and they experienced difficulties with normal conversation and their schoolwork. These children were classified as having a moderate disability (Table 3).

Twelve children assessed at age 7 years wore glasses (Table 8) and those with a correctable refractive error were recorded as having impairment. Two children, both ECMO, were registered blind or partially sighted and were classified with a severe and mild disability, respectively.

Overall Outcome

The overall outcome from trial entry to age 7 years of age is illustrated in Table 9. Additional information about the 18 children with moderate or severe disability is shown in Appendix 1.

The increased survival among those children randomized to a UK ECMO policy is not offset by disability among survivors. The relative risk for death or severe disability is 0.64 (95% CI 0.47–0.86, $P = .004$).

DISCUSSION

The UK Collaborative ECMO Trial has provided an opportunity to describe the implications of the treatment children receive when they experience severe respiratory failure in the newborn period. It is remarkable that many of these children, faced with life-defining disease, could be supported through this devastating illness and survive to lead enriching lives free of handicap. The trial has shown an ECMO policy to increase the likelihood of intact survival. The economic implications of this are reported elsewhere.¹³

The children involved in this study were recruited up to 10 years ago and hence it is valid to consider whether these data are still clinically relevant. Around the world, rates of neonatal ECMO referral have fallen, and use of interventions such as nitric oxide and high-frequency ventilation have grown. However, in general, trials of nitric oxide in term infants have shown that a majority have gone on to need ECMO. As a result, understanding the risk for long-term neurodevelopmental problems associated with this degree of illness in a mature newborn remains relevant to those who have to make decisions whether or when to use ECMO. Certainly, there are no similar data to suggest an advantage in terms of long-term outcome from avoiding ECMO once standard ECMO referral criteria have been reached.

This current work describes the difficulties some of children experience as a result of their early illness and the implication of these problems as they start school. At 7 years of age, a clearer profile of the learning difficulties and behavior problems is apparent than could be assessed at 4 years. The difficulties detectable at earlier assessment appear to be predictive of later problems. The 3 children who were classified with severe disability at 4 years are the same 3 children with severe disability at 7 years, and 15 of 16 children classified to have moderate disability at 4 years were also considered to have a disability when assessed at 7 years.

The specific patterns of learning difficulties identified in this study include a global cognitive loss, poor spatial skills, and difficulties with reading comprehension and visual and verbal memory. The assessment of word reading showed that the children's mean reading age was below average. In addition to this, however, the children's ability to interpret or process the information they had read was further compromised with respect to their reading age. The children were shown a complex diagram and asked to redraw the diagram at timed intervals through the assessment. They were observed to have difficulty organizing, a strategy for immediate recall of the design. This failure in processing skills combined with poor spatial abilities resulted in their poor scores in visual memory testing. These results suggest that the children are experiencing difficulties with both visual and spatial information processing. These problems might be expected to have an increasing effect on the

TABLE 7 Behavior at 7 Years

	Parent Reported		Teacher Reported	
	ECMO (n = 56)	CM (n = 34)	ECMO (n = 56)	CM (n = 34)
Behavior scale; no. (%) abnormal) (not known)				
Prosocial; no. (%) abnormal	4 (7) (2)	0 (0) (0)	5 (9) (4)	7 (21) (0)
Hyperactivity; no. (%) abnormal	14 (25) (2)	15 (44) (1)	13 (23) (5)	9 (27) (0)
Emotional; no. (%) abnormal	8 (14) (2)	12 (35) (0)	4 (7) (4)	6 (18) (0)
Conduct; no. (%) abnormal	12 (21) (2)	10 (29) (0)	5 (9) (4)	5 (15) (0)
Peer problems; no. (%) abnormal	9 (16) (3)	11 (32) (0)	5 (9) (5)	5 (15) (0)
Total deviance score (derived from hyperactivity, emotional, conduct, and peer problems scores); no. (%) abnormal)	10 (18) (3)	13 (38) (1)	7 (13) (5)	8 (24) (0)
Impact assessment score (behavioral difficulties reported to interfere with child's everyday life; score 0 = not at all, 1 = only a little, 2 = quite a lot, 3 = a great deal); no. (%)—score 2 or more	8 (15) (3)	9 (27) (1)	11 (21) (5)	6 (18) (0)

TABLE 8 Hearing and Vision at 7 Years

	ECMO (n = 56)	CM (n = 34)
Hearing; no. (%) (not known)		
Has hearing aids	2 (4) (2)	3 (9) (0)
Has ever had grommets inserted	5 (9) (2)	4 (12) (0)
Passed 20-dB sweep right and left	39 (70)	25 (74) (0)
Hearing loss any frequency in better ear		
1-dB loss	5 (9)	3 (9)
21-dB-dB loss	7 (13)	4 (12)
40-dB-dB loss	1 (2)	0 (0)
Not testable	3 (5) ^a	2 (6)
Not known	1 (2)	0 (0)
Isolated high-tone frequency 60-dB loss bilaterally 2000–4000 Hz (no aids)	0	0
Vision: no. (%) (not known)		
Wears glasses	6 (11) (2)	6 (18) (0)
Registered blind or partially blind	2 (4) (2)	0 (0)
Has had operation for squint	1 (2) (2)	0 (0)

^a Two children were not testable because their hearing aids were in place. One child was not able to comply with testing.

TABLE 9 Overall Outcome Classification of Children Recruited to the UK Collaborative ECMO Trial to Age 7 Years

Overall Status by 7 Years of Age	ECMO (N = 93) (%)	CM (N = 92) (%)
Deaths	31 (33)	54 (59)
Lost to follow-up	6 (6)	4 (4)
Children with:		
Severe disability	3 (3)	0
Moderate disability	9 (10)	6 (7)
Mild disability	13 (14)	11 (12)
Children with:		
Impairment only	21 (23)	15 (16)
No abnormal signs or disability	10 (11)	2 (2)
Assessed survivors with no disability	31/56 (55)	17/34 (50)

children's achievement and independent learning ability as they progress through school.

Sick neonates requiring respiratory support are at increased risk of sensorineural hearing loss (SNHL).¹⁵ Hearing impairment has been identified in 21 of 40 (53%) infants after treatment with hyperventilation for persistent fetal circulation.¹⁶ The extent of hearing loss was commonly worse in higher sound frequencies. Bilateral high-tone SNHL has also been identified in 3% to 21% (average 7.5%) of infants treated with neonatal ECMO.¹⁷ In both conventional and ECMO treatment, hearing impairment is often an isolated disability. It can be of late onset and progressive.^{18,19} SNHL has been positively correlated with the duration of ventilation, degree of alkalosis in infants treated conventionally, and similarly with a PaCO₂ below 14 mm Hg (1.35 kPa) before ECMO support.^{16,20}

In the past, there has been much debate²¹ about the etiology of the long-term developmental problems frequently observed in mature neonates who, in the neonatal period, develop cardiorespiratory problems of sufficient severity to merit ECMO. The focus of the discussion has been the extent to which the developmental problems are caused by the use of ECMO (in those that go on to get ECMO) with its inherent disruption of the cerebral circulation as opposed to the underlying cardiorespiratory instability being responsible. The data that have emerged from the various stages of the UK Collaborative ECMO Trial, and now confirmed quite clearly at age 7, suggest that the underlying disease process (and associated physiological instability) seems to be the major influence. Although both groups have had problems, the ECMO group has consistently performed better and with no evidence of an increase in problems that might be explained by unilateral brain lesions, which might be anticipated to be associated with cannulation. In summary, the beneficial influence of an ECMO policy is still present at the age of 7 years.

APPENDIX 1 Details of Survivors With Moderate or Severe Disability

Allocated to ECMO	Description of Outcome
Severe disability	
Male; identified with severe disability at 4 y (progressive since birth)	Could not carry out any tests; severely disabled in all domains except hearing, which scored moderately disabled; attends special school; requires multiple helpers; receives medical support through school; indwelling PEG for nutritional supplementation
Male; identified with severe disability at 4 y	Severely disabled in cognitive domain; moderately disabled in neuromotor, hearing, and vision domains; attends special school; requires continuous individual support in school; partially mobile in walking frame
Female; identified with severe disability at 4 y	Severely disabled in cognitive domain; moderately disabled in neuromotor and hearing domains; bilateral hearing aids; pragmatic and expressive deficits in language; attends special unit within mainstream school; requires continuous individual support in school
Moderate disability	
Male; identified with moderate disability at 4 y	Moderately disabled in cognitive and neuromotor domains; mildly disabled in behavioral domain; poor spatial ability; fine motor skills, and balance; hyperactivity reported
Male; identified with moderate disability at 4 y	Moderately disabled in behavioral domain; school progress normal; hyperactivity reported; learning support provided at school
Male; identified with moderate disability at 4 y	Moderately disabled in cognitive domain; poor spatial ability; poor social skills and emotional problems reported; school progress below age norms; requires full-time learning support at school
Female; identified with moderate disability at 4 y	Moderately disabled in cognitive and hearing domains; receives weekly speech therapy at school
Male; identified with moderate disability at 4 y	Moderately disabled in cognitive and behavioral domains
Male; identified with mild disability at 4 y	Moderately disabled in neuromotor and behavior domains; mildly disabled in cognitive domain; hyperactivity and emotional problems reported; has learning support at school
Male; identified with mild disability at 4 y	Moderately disabled in behavior domain; hyperactivity and conduct disorder reported
Male; identified with moderate disability at 4 y	Moderately disabled in cognitive domain; poor spatial ability; school progress below age norms; requires full-time learning support at school
Male; identified with moderate disability at 4 y	Moderately disabled in cognitive domain; mildly disabled in neuromotor domain; poor fine motor skills; school progress below age norms; requires full-time learning support at school

ACKNOWLEDGMENTS

We thank the infants and their parents who took part in the trial; their General Practitioners, Health Visitors, Head Teachers, and Teachers; all of the doctors, nurses, and their colleagues in the participating centers; and Ann Johnson, Carole Harris, Sarah Ayers, and Stavros Petrou at the National Perinatal Epidemiology Unit. The trial was funded by the England and Wales Department of Health and the Scottish Office Department of Health.

REFERENCES

- UK Collaborative Trial Group. UK collaborative randomised trial of neonatal extracorporeal membrane oxygenation. *Lancet*. 1996;348:75–82
- UK Collaborative ECMO Group. The Collaborative ECMO Trial: follow-up to 1 year of age [Abstract]. *Pediatrics*. 1998;101:690. Available at: www.pediatrics.org/cgi/content/full/101/4/e1
- Roberts TE on behalf of the UK Collaborative ECMO Group. Economic evaluation and randomised controlled trial of extracorporeal membrane oxygenation: UK Collaborative ECMO Trial. *BMJ*. 1998;317:911–915
- Elbourne D, Field D, Mugford M. Extracorporeal membrane oxygenation for severe respiratory failure in newborn infants. *Cochrane Database Syst Rev*. 2002;1:CD001340
- Goldman R. Strengths and Difficulties Questionnaire. *J Child Psychol Psychiatry*. 1997;38:581–586
- Bishop DVM. Development of the Children's Communication Checklist (CCC). A method for assessing qualitative aspects of communicative impairment in children. *J Child Psychol Psychiatry*. 1998;39:879–891
- Goldberg D, Williams P. *General Health Questionnaire*. Windsor, United Kingdom: NFER-Nelson Publishing Co; 1991
- British Ability Scales*. 2nd ed. (BAS II). Windsor, United Kingdom: NFER-Nelson Publishing Co; 1996
- Rust J, Golombok S, Trickey G. *Wechsler Objective Reading Dimensions (WORD)*. London, United Kingdom: The Psychological Corp; 1993
- Meyes J, Meyes K. *Rey Complex Figure Test*. Odessa, FL: Psychological Resources; 1995
- Cohen MJ. *Children's Memory Scale*. London, United Kingdom: Psychological Corp; 1992
- Henderson SE, Sugden DA. *The Movement Assessment Battery for Children*. London, United Kingdom: Psychological Corp; 1992
- Petrou S, Bischof M, McNally H. Cost-effectiveness of neonatal ECMO based on 7-year results from the UK Collaborative ECMO Trial. *Pediatrics*. 2006;117:1641–1651
- Bennett CC, Johnson A, Field D, Elbourne D. UK collaborative randomised trial of neonatal extracorporeal membrane oxygenation: follow-up to age 4 years. *Lancet*. 2001;357:1094–1096
- McClelland RJ, Watson DR, Lawless V, Houston HG, Adams D. Reliability and effectiveness of screening for hearing loss in high-risk neonates. *BMJ*. 1992;304:806–809
- Hendricks-Munoz KD, Walton JP. Hearing loss in infants with persistent fetal circulation. *Pediatrics*. 1988;81:5:650–656
- Cheung PY, Robertson CM. Sensorineural hearing loss in survivors of neonatal extracorporeal membrane oxygenation. *Pediatr Rehabil*. 1997;1:127–130
- Kwashiro N, Tsuchihashi N, Koga K, Itoh Y. Delayed post-natal intensive care hearing disturbance. *Int J Pediatr Otorhinolaryngol*. 1996;34:35–43
- Lasky RE, Wioreck L, Becker TR. Hearing loss in survivors of neonatal ECMO therapy and HFO therapy. *Journal of American Audiology*. 1998;9:164–173
- Graziani LJ, Gringlas M, Baumgart S. Cerebrovascular complications and neurodevelopmental sequelae of neonatal ECMO. *Clin Perinatol*. 1997;24:655–675
- Pearson GA, Field DJ, Firmin RK, Sosnowski AS. UK experience in neonatal extracorporeal membrane oxygenation. *Arch Dis Child*. 1992;67:822–825

**United Kingdom Collaborative Randomized Trial of Neonatal Extracorporeal
Membrane Oxygenation: Follow-up to Age 7 Years**

Helena McNally, Charlotte C. Bennett, Diana Elbourne, David J. Field and for the UK
Collaborative ECMO Trial Group

Pediatrics 2006;117:e845-e854; originally published online Apr 24, 2006;

DOI: 10.1542/peds.2005-1167

Updated Information & Services	including high-resolution figures, can be found at: http://www.pediatrics.org/cgi/content/full/117/5/e845
References	This article cites 14 articles, 3 of which you can access for free at: http://www.pediatrics.org/cgi/content/full/117/5/e845#BIBL
Citations	This article has been cited by 2 HighWire-hosted articles: http://www.pediatrics.org/cgi/content/full/117/5/e845#otherarticles
Subspecialty Collections	This article, along with others on similar topics, appears in the following collection(s): Premature & Newborn http://www.pediatrics.org/cgi/collection/premature_and_newborn
Permissions & Licensing	Information about reproducing this article in parts (figures, tables) or in its entirety can be found online at: http://www.pediatrics.org/misc/Permissions.shtml
Reprints	Information about ordering reprints can be found online: http://www.pediatrics.org/misc/reprints.shtml

American Academy of Pediatrics

DEDICATED TO THE HEALTH OF ALL CHILDREN™

